

Variable List for Causality Testing

ID	Variable Name	Level(s)	Time
1	TMP Temperature [K]	200 mb	Analysis
2	VVEL Vertical Velocity (Pressure) [Pa/s]	200 mb	Analysis
3	UGRD U-Component of Wind [m/s]	200 mb	Analysis
4	VGRD V-Component of Wind [m/s]	200 mb	Analysis
5	TMP Temperature [K]	300 mb	Analysis
6	DPT Dew Point Temperature [K]	300 mb	Analysis
7	VVEL Vertical Velocity (Pressure) [Pa/s]	300 mb	Analysis
8	UGRD U-Component of Wind [m/s]	300 mb	Analysis
9	VGRD V-Component of Wind [m/s]	300 mb	Analysis
10	TMP Temperature [K]	400 mb	Analysis
11	DPT Dew Point Temperature [K]	400 mb	Analysis
12	VVEL Vertical Velocity (Pressure) [Pa/s]	400 mb	Analysis
13	UGRD U-Component of Wind [m/s]	400 mb	Analysis
14	VGRD V-Component of Wind [m/s]	400 mb	Analysis
15	TMP Temperature [K]	500 mb	Analysis
16	DPT Dew Point Temperature [K]	500 mb	Analysis
17	VVEL Vertical Velocity (Pressure) [Pa/s]	500 mb	Analysis
18	UGRD U-Component of Wind [m/s]	500 mb	Analysis
19	VGRD V-Component of Wind [m/s]	500 mb	Analysis
20	TMP Temperature [K]	600 mb	Analysis
21	DPT Dew Point Temperature [K]	600 mb	Analysis
22	VVEL Vertical Velocity (Pressure) [Pa/s]	600 mb	Analysis
23	UGRD U-Component of Wind [m/s]	600 mb	Analysis
24	VGRD V-Component of Wind [m/s]	600 mb	Analysis
25	TMP Temperature [K]	700 mb	Analysis
26	DPT Dew Point Temperature [K]	700 mb	Analysis
27	VVEL Vertical Velocity (Pressure) [Pa/s]	700 mb	Analysis
28	UGRD U-Component of Wind [m/s]	700 mb	Analysis
29	VGRD V-Component of Wind [m/s]	700 mb	Analysis
30	TMP Temperature [K]	775 mb	Analysis
31	DPT Dew Point Temperature [K]	775 mb	Analysis
32	VVEL Vertical Velocity (Pressure) [Pa/s]	775 mb	Analysis

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ID	Variable Name	Level(s)	Time
33	UGRD U-Component of Wind [m/s]	775 mb	Analysis
34	VGRD V-Component of Wind [m/s]	775 mb	Analysis
35	TMP Temperature [K]	850 mb	Analysis
36	RH Relative Humidity [%]	850 mb	Analysis
37	DPT Dew Point Temperature [K]	850 mb	Analysis
38	VVEL Vertical Velocity (Pressure) [Pa/s]	850 mb	Analysis
39	UGRD U-Component of Wind [m/s]	850 mb	Analysis
40	VGRD V-Component of Wind [m/s]	850 mb	Analysis
41	TMP Temperature [K]	925 mb	Analysis
42	RH Relative Humidity [%]	925 mb	Analysis
43	DPT Dew Point Temperature [K]	925 mb	Analysis
44	UGRD U-Component of Wind [m/s]	925 mb	Analysis
45	VGRD V-Component of Wind [m/s]	925 mb	Analysis
46	ABSV Absolute Vorticity [1/s]	925 mb	Analysis
47	TMP Temperature [K]	1000 mb	Analysis
48	DPT Dew Point Temperature [K]	1000 mb	Analysis
49	VVEL Vertical Velocity (Pressure) [Pa/s]	1000 mb	Analysis
50	UGRD U-Component of Wind [m/s]	1000 mb	Analysis
51	VGRD V-Component of Wind [m/s]	1000 mb	Analysis
52	ABSV Absolute Vorticity [1/s]	1000 mb	Analysis
53	TSOIL Soil Temperature [K]	0-0 m below ground	Analysis
54	SOILW Volumetric Soil Moisture Content [Fraction]	0-0 m below ground	Analysis
55	TSOIL Soil Temperature [K]	0.01-0.01 m below ground	Analysis
56	SOILW Volumetric Soil Moisture Content [Fraction]	0.01-0.01 m below ground	Analysis
57	TSOIL Soil Temperature [K]	0.04-0.04 m below ground	Analysis
58	SOILW Volumetric Soil Moisture Content [Fraction]	0.04-0.04 m below ground	Analysis
59	TSOIL Soil Temperature [K]	0.1-0.1 m below ground	Analysis
60	SOILW Volumetric Soil Moisture Content [Fraction]	0.1-0.1 m below ground	Analysis
61	TSOIL Soil Temperature [K]	0.3-0.3 m below ground	Analysis
62	SOILW Volumetric Soil Moisture Content [Fraction]	0.3-0.3 m below ground	Analysis
63	TSOIL Soil Temperature [K]	0.6-0.6 m below ground	Analysis
64	SOILW Volumetric Soil Moisture Content [Fraction]	0.6-0.6 m below ground	Analysis

Variable List for Causality Testing

ID	Variable Name	Level(s)	Time
65	MSTAV Moisture Availability [%]	0 m underground	Analysis
66	REFC Composite reflectivity [dB]	entire atmosphere	Analysis
67	VIL Vertically-Integrated Liquid Water [kg/m ²]	entire atmosphere	Analysis
68	GUST Wind Speed (Gust) [m/s]	surface	Analysis
69	DZDT Vertical Velocity (Geometric) [m/s]	0.5-0.8 sigma layer	0-0 day ave fcst
70	TCOLG Total Column Integrate Graupel [kg/m ²]	entire atmosphere (as one layer)	0-0 day max fcst
71	LTNG Lightning [non-dim]	surface	Analysis
72	PRES Pressure [Pa]	surface	Analysis
73	TMP Temperature [K]	surface	Analysis
74	TMP Temperature [K]	2 m above ground	Analysis
75	SPFH Specific Humidity [kg/kg]	2 m above ground	Analysis
76	DPT Dew Point Temperature [K]	2 m above ground	Analysis
77	RETOP Echo Top [m]	cloud top	1 hour fcst
78	MAXUVV Hourly Maximum of Upward Vertical Velocity [m/s]	100-1000 mb above ground	0-1 hour max fcst
79	MAXDVV Hourly Maximum of Downward Vertical Velocity [m/s]	100-1000 mb above ground	0-1 hour max fcst
80	UGRD U-Component of Wind [m/s]	80 m above ground	1 hour fcst
81	VGRD V-Component of Wind [m/s]	80 m above ground	1 hour fcst
82	UGRD U-Component of Wind [m/s]	10 m above ground	1 hour fcst
83	VGRD V-Component of Wind [m/s]	10 m above ground	1 hour fcst
84	WIND Wind Speed [m/s]	10 m above ground	0-1 hour max fcst
85	PRATE Precipitation Rate [kg/m ² /s]	surface	1 hour fcst
86	APCP Accumulated Precipitation [mm]	surface	0-1 hour acc fcst
87	LFTX Surface Lifted Index [K]	500-1000 mb	1 hour fcst
88	CAPE Convective Available Potential Energy [J/kg]	surface	1 hour fcst
89	CIN Convective Inhibition [J/kg]	surface	1 hour fcst
90	PWAT Precipitable Water [kg/m ²]	entire atmosphere (a one layer)	1 hour fcst
91	LCDC Low Cloud Cover [%]	low cloud layer	1 hour fcst
92	MCDC Medium Cloud Cover [%]	middle cloud layer	1 hour fcst
93	HCDC High Cloud Cover [%]	high cloud layer	1 hour fcst
94	TCDC Total Cloud Cover [%]	entire atmosphere	1 hour fcst
95	DSWRF Downward Short-Wave Radiation Flux [W/m ²]	surface	1 hour fcst
96	HLCY Storm Relative Helicity [m ² /s ²]	3000-0 m above ground	1 hour fcst

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ID	Variable Name	Level(s)	Time
97	HLCY Storm Relative Helicity [m^2/s^2]	1000-0 m above ground	1 hour fcst
98	USTM U-Component Storm Motion [m/s]	0-6000 m above ground	1 hour fcst
99	VSTM V-Component Storm Motion [m/s]	0-6000 m above ground	1 hour fcst
100	VUCSH Vertical U-Component Shear [$1/\text{s}$]	0-1000 m above ground	1 hour fcst
101	VVCSH Vertical V-Component Shear [$1/\text{s}$]	0-1000 m above ground	1 hour fcst
102	VUCSH Vertical U-Component Shear [$1/\text{s}$]	0-6000 m above ground	1 hour fcst
103	VVCSH Vertical V-Component Shear [$1/\text{s}$]	0-6000 m above ground	1 hour fcst
104	CAPE Convective Available Potential Energy [J/kg]	180-0 mb above ground	1 hour fcst
105	CIN Convective Inhibition [J/kg]	180-0 mb above ground	1 hour fcst
106	HPBL Planetary Boundary Layer Height [m]	surface	1 hour fcst
107	CAPE Convective Available Potential Energy [J/kg]	90-0 mb above ground	1 hour fcst
108	CIN Convective Inhibition [J/kg]	90-0 mb above ground	1 hour fcst
109	CAPE Convective Available Potential Energy [J/kg]	255-0 mb above ground	1 hour fcst
110	CIN Convective Inhibition [J/kg]	255-0 mb above ground	1 hour fcst
111	LTNG Lightning [non-dim]	entire atmosphere	Analysis
112	CNWAT Plant Canopy Surface Water [kg/m^2]	surface	1 hour fcst
113	RH Relative Humidity [%]	2 m above ground	1 hour fcst
114	SFCR Surface Roughness [m]	surface	1 hour fcst
115	FRICV Frictional Velocity [m/s]	surface	1 hour fcst
116	SHTFL Sensible Heat Flux [W/m^2]	surface	1 hour fcst
117	LHTFL Latent Heat Flux [W/m^2]	surface	1 hour fcst
118	GFLUX Ground Heat Flux [W/m^2]	surface	1 hour fcst
119	DLWRF Downward Long-Wave Radiation Flux [W/m^2]	surface	1 hour fcst
120	USWRF Upward Short-Wave Radiation Flux [W/m^2]	surface	1 hour fcst
121	ULWRF Upward Long-Wave Radiation Flux [W/m^2]	surface	1 hour fcst
122	PRES Pressure [Pa]	0C isotherm	1 hour fcst
123	N/A		
124	N/A		
125	N/A		
126	N/A		
127	N/A		
128	N/A		